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Wave propagation

TFE4130 Exercise 5

Department of Electronic systems

1. Special angles

a) What is total reflection? How can total reflection be utilized in a waveguide?

b) Use Snell's law to find the critical angle θ_c exprienced by a wave propagating in silica $n_1 = 1.44$ in the boundary between the material and air, $n_1 > n_2 = n_{\text{air}}$.

c) What is the Brewster angle?

2. Coaxial cable

A coaxial cable consists of an inner conductor of radius a, and an outer conductor, whose inner radius is b, and thickness is negligible. Between the conductors there is a medium with permittivity ϵ and permeability μ. We assume a negligible resistance in the conductors. A voltage V delivers a current I through the cable, to a load R_{L}. Assume DC-conditions (i.e. the current in the inner conductor is evenly distributed over its cross section), and that the cable is infinitely long.

a) Show that the **E**-field over the cross-section of the cable for the situations $a < r < b$ decays like $1/r$. You can use Gauss' law and integrate over a fragment of the cable as shown in Figure 1:

$$Q_{\text{in } S} = \int_S \mathbf{D} \cdot d\mathbf{S} = l \int_0^{2\pi} rD(r) d\theta. \tag{1}$$

You will then find an expression for $D(r)$ that you can plug into $E(r) = D(r)/\epsilon$.

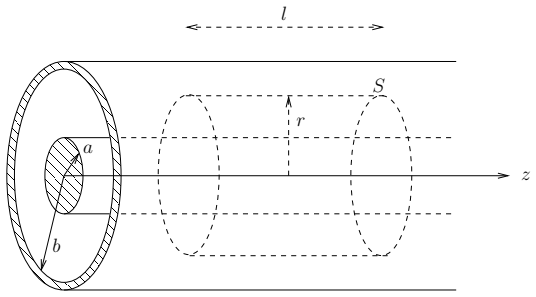


Figure 1: The coaxial cable with a suggested surface integration by Gauss' law.

b) Explain why the **E**-field is zero for $r < a$ and $r > b$.

c) Calculate the **H**-field over the cross-section of the cable. Look at the situations $r < a$, $a < r < b$, and $r > b$. Consider symmetry to simplify your calculations. You may use

Ampere's law and integrate along a closed loop enclosing the electric current I as suggested in Figure 2:

$$I_{\text{through C}} = I\frac{r^2}{a^2} \implies H(r) = \frac{Ir}{2\pi a^2}. \tag{2}$$

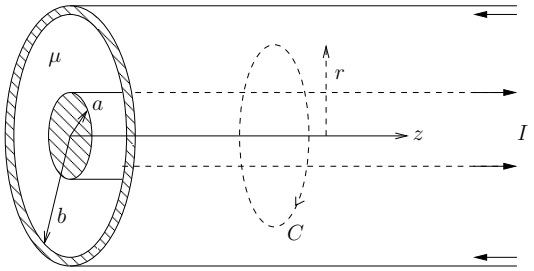


Figure 2: The coaxial cable with a suggested closed loop integration by Ampere's law.

d) Calculate the Poynting's vector for the situations $r < a$, $a < r < b$, and $r > b$. Remember that Poynting's vector is $\mathcal{P} = \mathbf{E} \times \mathbf{H}$. We have that $\mathbf{E} \perp \mathbf{H}$, so $\mathcal{P} = EH$.

e) Sketch the direction of the electric and magnetic fields, and the Poynting vector in relation to the coaxial cable.

3. Transmission line

Figure 3 shows a transmission line with $L = 0.18$ μH/m, $C = 50$ pF/m, $Z_{\text{L}} = 75$ Ω and characteristic impedance Z_0.

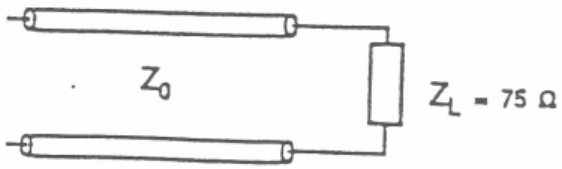


Figure 3: Transmission line, problem 1.

a) Find the characteristic impedance Z_0. Explain what is meant by "the characteristic impedance". Does Z_0 depend on the length of the line?

b) Find the reflection coefficient Γ (named Γ_L in the lectures). Where on the line is Γ defined?

c) Find the standing-wave ratio.

d) Find Γ when $Z_{\mathrm{L}} = 0$.

e) Find Γ when $Z_{\mathrm{L}} \to \infty$.

f) Find Γ when $Z_{\mathrm{L}} = Z_0$.